

Trackformer 1.2: Coupled Pressure-Field Evolution for Tropical-Cyclone Forecasting

Architecture, Forecast Dynamics and Development Evaluation

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<https://github.com/yu314-coder/typhoon-predict>

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Abstract

Trackformer 1.2 advances an atmospheric basin state and a translating pressure core to forecast tropical-cyclone motion and central pressure. Recurrent convolutional networks evolve the fields; multiscale attention supplies environmental context. The reported centre is associated with a depression in the predicted core, and central pressure is sampled from that same field. Twenty six-hour transitions produce a five-day forecast. We describe the released architecture, input contract, update equations, objectives and 50-member input-perturbation ensemble, illustrated by a module diagram. On a previously inspected development cohort of 270 forecast cases from 90 storms, mean track error is 714.4 km for the 1.2 ensemble versus 902.3 km for the archived 1.1 route pipeline. These are output comparisons, not a controlled architecture ablation or evidence of untouched generalization. A selected Fung-wong example illustrates geographic route and pressure comparisons while retaining visible position, timing and intensity differences. All figures retain the saved predictions and explicitly separate aggregate evidence from selected examples.

1 Design and forecast state

The system is not a route predictor followed by an independently drawn pressure image. It evolves pressure and atmospheric states, proposes a translation of a storm-following frame, then associates a centre with the predicted pressure within that frame. Translation is learned and steering-conditioned: this is neither a pressure-only tracker nor a complete atmospheric numerical solver.

At step k , the recurrent state is

$$s_k = (g_k, h_k^g, c_k, h_k^c, a_k, o_k, z_k, m_k, v_k; G, R, I, M).$$

Here g is the normalized eight-channel basin field; h^g its memory; c the core pressure; h^c core memory; a its anomaly relative to the basin; o the core-grid origin; z the associated storm centre; m displacement per six hours; and v auxiliary maximum wind. Static geography G , regional coordinates R , issue intensity I and masks M stay fixed.

2 Inputs, geometry and outputs

Nine past analyses at $-48, -42, \dots, 0$ h form a tensor of shape $B \times 9 \times 8 \times 25 \times 33$. The basin spans 60 to 0°N and 100 to 180°E at 2.5° spacing. Channels are

$$[\text{MSLP}, Z_{500}, u_{850}, v_{850}, u_{500}, v_{500}, u_{200}, v_{200}],$$

in hPa, metres and metres per second. Channel normalization is $\tilde{x}_j = (x_j - \mu_j)/\sigma_j$, with the released training-only constants. Statistics are not refitted for each forecast.

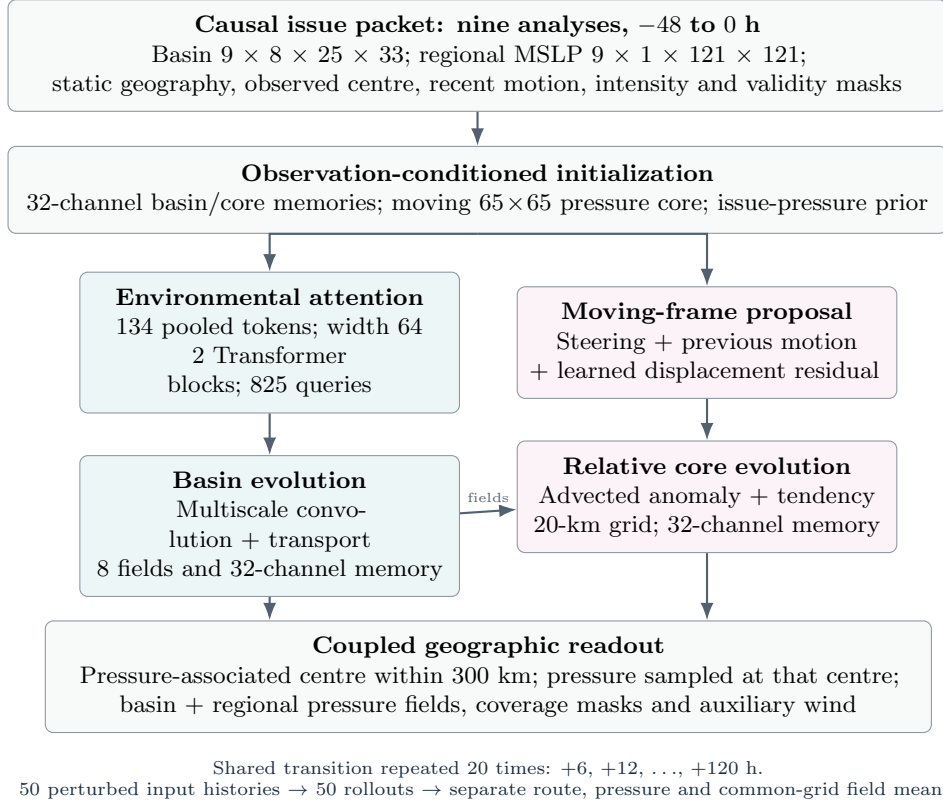


Figure 1: Released model structure. The basin branch supplies environmental information to the moving core; motion is proposed before the pressure core determines the reported centre. Boxes show functional groups, not independent models. Future observations are excluded from inference. The ensemble contains 50 input realizations of one network, not 50 trained networks.

Optional regional pressure history has shape $B \times 9 \times 1 \times 121 \times 121$ on an issue-anchored patch sampled every 0.25° , with offsets of $\pm 15^\circ$. A detail-availability flag chooses native regional history or coarse basin pressure during initialization. The regional coordinate grid is still supplied when native observations are unavailable.

Four static channels encode latitude/90, longitude/180 – 1, land fraction and elevation/8000 m. The issue state supplies latitude/longitude, east/north motion in km per six hours, maximum wind in knots, central pressure in hPa and two intensity-availability masks. Only observations and analyses at or before issue time may enter inference; retrospective data availability is a separate operational question.

The internal core is a 65×65 grid at 20-km sampling over ± 640 km. Its field is a learned reconstruction, not native 20-km observations. For each of twenty leads (6, 12, $\dots$, 120 h), outputs include basin fields, regional pressure, core pressure/coordinates, storm centre, central pressure, domain masks and auxiliary maximum wind. This implementation does not output quadrant wind radii or a resolved surface-wind map.

3 Neural modules

The released non-tiny architecture has 21,452,595 trainable parameters.

3.1 Multiresolution evolution networks

The basin network receives 44 channels: eight fields, 32 memory features and four static channels. Encoder widths are 72, 144, 288, 432; residual-block depths are 2, 2, 4, 4. A 3×3 stem and stride-two convolutions produce four scales. The decoder uses 1×1 projections, bilinear upsampling, additive skip connections and one residual block at each reconstructed scale. Output heads emit ten transition channels and 32 recurrent features. Eight transition channels are field tendencies and two are flow corrections.

Each residual block comprises a depthwise 5×5 convolution, eight-group normalization, 1×1 expansion by four, GELU and 1×1 contraction. A learned residual multiplier starts at 0.01. Transition heads start at zero. The core network uses widths 64, 128, 256, 384 and the same depths. Its 45 inputs are core pressure, eight environmental fields, 32 core-memory channels and four geographic channels. Three outputs represent one anomaly tendency and two local-flow corrections.

3.2 Convolutional recurrent memory

Basin and core memories each have 32 channels. History encoders receive 72 flattened basin channels or nine pressure-history channels, respectively. Given advected memory h and new evolution features f ,

$$\begin{aligned}(r, u) &= \text{sigmoid}(W_g * [h, f]), \\ q &= \tanh(W_q * [r \odot h, f]), \\ h' &= (1 - u) \odot h + u \odot q.\end{aligned}$$

The gate convolution emits 64 channels split into reset and update gates; the candidate convolution emits 32. Forecast leads share these recurrent modules rather than using twenty independent predictors.

3.3 Environmental attention

A 1×1 convolution projects $[g, h^g, G]$ from 44 to 64 channels. Average pooling at sizes/strides three and five with ceiling mode yields $9 \times 11 + 5 \times 7 = 134$ environmental tokens. Two pre-normalization Transformer layers use width 64, four heads, feed-forward width 128, ReLU activation and zero dropout. Each of the 825 basin cells supplies a query projected from its memory to width 64. Four-head cross-attention retrieves context; a zero-initialized $64 \rightarrow 32$ projection adds it to basin memory before the transition. No separate positional embedding is added: geographic coordinates are already among the token inputs.

4 Observation-conditioned initialization

Write $\mathcal{S}(F, \xi)$ for bilinear field sampling. Sampling uses FP32 geometry, aligned corners and border padding for the basin; transported core anomalies use zero padding. All nine pressure histories are sampled onto the core grid centred at the issue observation. A detail mask chooses native pressure or coarse pressure. The latest map initializes c_0 , while all nine maps initialize the core memory.

If issue central pressure is valid, the initial core is corrected by

$$c_0(x, y) \leftarrow c_0(x, y) + (\tilde{p}_{\text{issue}} - c_0(0, 0))e^{-(x^2+y^2)/(2 \cdot 100^2)}T(x, y),$$

where x, y are kilometres and $T = \text{clip}((640 - \sqrt{x^2 + y^2})/160, 0, 1)$. The anomaly is $a_0 = c_0 - \mathcal{S}(g_{0, \text{MSLP}}, \xi_0)$. This compact prior is applied only at initialization. Future scalar pressure is not repeatedly inserted into the field.

5 One six-hour transition

5.1 Basin transport

Learned softmax weights blend physical winds at 850, 500 and 200 hPa. Their initialization is $(0.4, 0.4, 0.2)$, not a fixed inference constraint. For network output d ,

$$U_k = \sum_{\ell} \alpha_{\ell}(u_{\ell}, v_{\ell})_k + 15 \tanh(d_{8:10}) \text{ m s}^{-1}.$$

Backtracing constructs departure coordinates $D(U_k)$ and

$$g_{k+1} = \mathcal{S}(g_k, D(U_k)) + 0.2 \tanh(d_{0:8}).$$

Six hours convert 1 m/s to 21.6 km. A latitude degree is treated as 111.2 km. A longitude degree is treated as $111.2 \max(\cos \phi, 0.2)$ km. The attention-conditioned memory is advected by the same departure grid and updated with new features. Tendencies are in normalized channel units. This semi-Lagrangian-style update is not a complete or exactly conservative atmospheric equation solver.

5.2 Moving-frame proposal

A $44 \rightarrow 64 \rightarrow 3$ GELU MLP reads sampled 32-channel memory, eight basin fields, previous motion divided by 100, and two issue-intensity masks. For output b ,

$$m_{k+1} = 0.7(21.6)U_k(z_k) + 0.3m_k + 100 \tanh(b_{0:2}).$$

This displacement, in km per six hours, defines a new origin relative to z_k . It proposes where the pressure core moves, but is not the final reported centre. The third output updates auxiliary wind: $v_{k+1} = \max(0, v_k + 5b_2)$.

5.3 Relative core transport

The core network reads the previous pressure core, its old sampled environment, core memory and geography at the proposed origin. Local velocity is $U^{\text{local}} = \mathcal{S}(U_k) + 5 \tanh(d_{1:3}^c)$. Subtracting $m_{k+1}/21.6$ removes bulk frame translation. Backward coordinates also account for the offset between the previous associated centre and its previous grid origin. On this relative departure grid D_c ,

$$\begin{aligned} a_{k+1} &= \mathcal{S}(a_k, D_c) + 0.1d_0^c, \\ c_{k+1} &= \mathcal{S}(g_{k+1, \text{MSLP}}, \xi_{k+1}) + Ta_{k+1}. \end{aligned}$$

Unlike the basin tendency, d_0^c is not bounded by $\tanh$. Core memory uses the same relative sampling followed by its gated update.

5.4 Centre association and central pressure

Convert core pressure to hPa: $P = \sigma_p c_{k+1} + \mu_p$. For cells within 300 km of the proposed origin,

$$w_i = \text{softmax}_i \left(-P_i/2 - r_i^2/(2 \cdot 180^2) \right).$$

Cells beyond 300 km are excluded. The weighted mean of cell latitude/longitude is z_{k+1} . Central pressure is the bilinear sample $\mathcal{S}(P, z_{k+1})$, not necessarily the grid minimum. The spatial prior associates the forecast with this storm rather than an unrelated remote depression.

5.5 Geographic pressure output

The issue-anchored regional output is

$$\hat{p}_R = \mathcal{S}(g_{k+1, \text{MSLP}}, R) + \mathcal{S}(Ta_{k+1}, R; \text{zero padding}),$$

then de-normalized to hPa. This basin-plus-core reconstruction is generated by the model itself; future observed fields and display-only vortices are not substituted. Its 0.25° sampling is not proof of native resolution. Masks indicate valid basin coverage. Resampling and localized association mean the reported centre, central pressure and regional map minimum need not coincide exactly. A display must preserve those distinctions rather than relocate fields.

6 Autoregression and learning objectives

Inference initializes once and repeats the transition twenty times, using only fixed issue context and its own predicted future state. The implemented single-step loss is

$$\begin{aligned} L = & L_{\text{env}} + L_{\text{regional}} + 0.2L_{\text{route}} + 0.5L_{\text{pressure}} + 0.05L_{\text{wind}} \\ & + 0.25L_{\Delta p} + 0.1L_{\Delta z}. \end{aligned}$$

The final terms require valid preceding-target labels. Environment loss is cosine-latitude-weighted Smooth L1 in normalized units. Regional loss includes Smooth L1 and 0.2 times adjacent-grid gradient errors on each axis. Absolute regional-field supervision excludes the inner 150 km around the target centre, because reanalysis can underresolve observed central pressure. This is a supervision choice, not proof of correct inner-core structure.

Route residuals use a 100-km scale; pressure and wind use 20 hPa and 30 kt. Pressure-change error uses 10 hPa and displacement-change error uses 100 km. All objectives use explicit masks. They train field evolution and readout together, without adding an independent central-pressure output head. Training-only normalization, storm-level split boundaries and causal input selection remain essential implementation constraints.

7 Single forecasts and 50-member means

Evaluation-mode inference is deterministic for fixed inputs and weights. The documented ensemble perturbs only normalized historical weather: independent Gaussian fields are spatially smoothed with kernels five (basin) and nine (regional), centred, unit-scaled per map and multiplied by amplitudes 0.025 and 0.015. Issue centre, motion, intensity, masks and geography remain fixed. Each member performs a complete recurrent rollout.

With $M = 50$, routes and central pressures are averaged separately: $\bar{z} = M^{-1} \sum_m z^{(m)}$ and $\bar{p} = M^{-1} \sum_m p^{(m)}$. Physical pressure fields are averaged on common geographic grids, not by averaging unaligned moving-core array indices. In general,

$$\bar{z} \neq \arg \min_x \bar{P}(x), \quad \bar{p} \neq \min_x \bar{P}(x).$$

These are perturbed-input realizations of one model, not 50 trained networks or learned latent samples. Spread is not automatically calibrated uncertainty. Member counts, perturbation settings and seed policy must accompany the outputs.

8 Source identity and use boundary

The public name is **Trackformer 1.2**. Its manifest retains the internal checkpoint identity, 1.2.73 epoch 4, and exact weight/source hashes. `model.py` implements environmental attention and added temporal losses; `baseline_model.py` implements moving-core evolution and readout; `v165_base.py` supplies sampling, evolution blocks and memories. All are in `models/trackformer_1.2_field/`.

Learned reconstructions, limited domain, input-source differences and retrospective availability constrain interpretation. The model remains a research artifact, not an operational warning system.

9 Development evaluation

9.1 Cohort and comparison boundary

The completed comparison comprises **270 issue cases from 90 storms**, with three issues per storm and twenty six-hour leads per case. The 1.2 forecast is the mean of 50 causal historical-input perturbations; 1.1 uses its archived causal route pipeline. Cases, labels and forecast leads are matched, but the input pipelines differ. This is not a controlled architecture ablation. The cohort was repeatedly inspected and includes cases outside the intended Western Pacific domain. It is therefore development evidence, not an untouched or basin-specific validation claim.

The original 1.1 archive stored absolute latitude/longitude under a local-km array name. The reproduced comparison converts those coordinates into the same issue-centred local projection used by 1.2 and checks a round trip to the original coordinates. Its corrected 902.3-km mean replaces the superseded 1,036.3-km score; original forecasts are preserved.

9.2 Position, direction and shape are different quantities

Mean local-coordinate position error improves by 20.8%, from 902.3 to 714.4 km; +120 h error is 1,833.2 versus 1,433.4 km. The +6 h error is *slightly worse* for 1.2 (74.2 versus 72.8 km), so the aggregate is not an improvement at every lead. Mean heading error is 56.22 versus 44.82 degrees. Centred route-shape similarity is 0.7273 versus 0.8249; normalized Fréchet path similarity is 0.4983 versus 0.5727. These metrics are not accuracy percentages and are not combined into a weighted rank.

Actual route alignment is the primary geographic criterion: predicted and observed positions should coincide at the same valid times. The shape score removes translation and scale, so it can remain high for a geographically displaced forecast. It must not substitute for position error or an unshifted route overlay. The path score responds to displacement but is not a strict same-time distance.

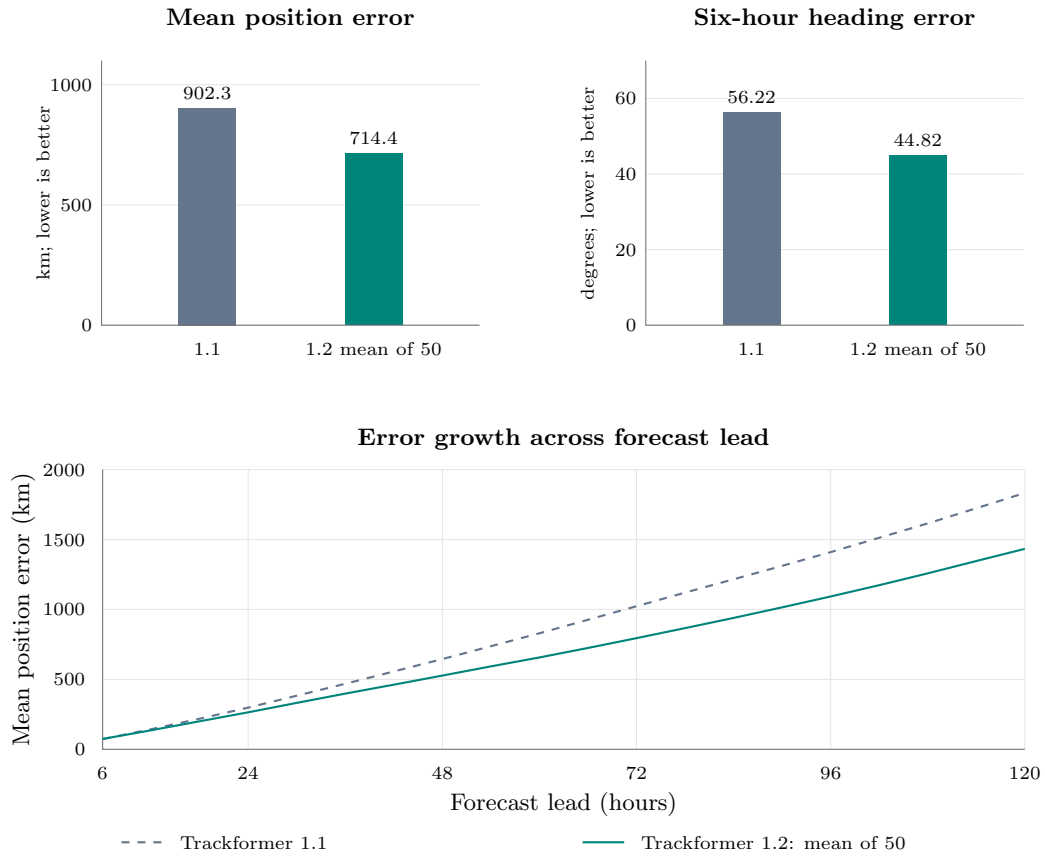


Figure 2: Matched development comparison. Top: all-lead position and heading errors. Bottom: mean position error at each lead. Distances use the saved local-km projection. Heading uses 5,382 common valid steps where truth and both forecasts move more than 1 km. All 270 cases remain in the position scores, including poor and out-of-domain cases. No uncertainty intervals are available in these saved results.

10 Selected Fung-wong case: alignment and intensity

The Fung-wong forecast is issued at 00 UTC on 7 November 2025, using the same saved 50-member policy. Figure 3 overlays predictions and exact six-hour observed coordinates without translating, rotating or scaling either route. Mean great-circle error is 130.9 km, +120 h error is 142.3 km and the maximum is 198.3 km: all twenty leads are within 200 km. These geographic errors are a separate diagnostic from the preceding local-coordinate scores.

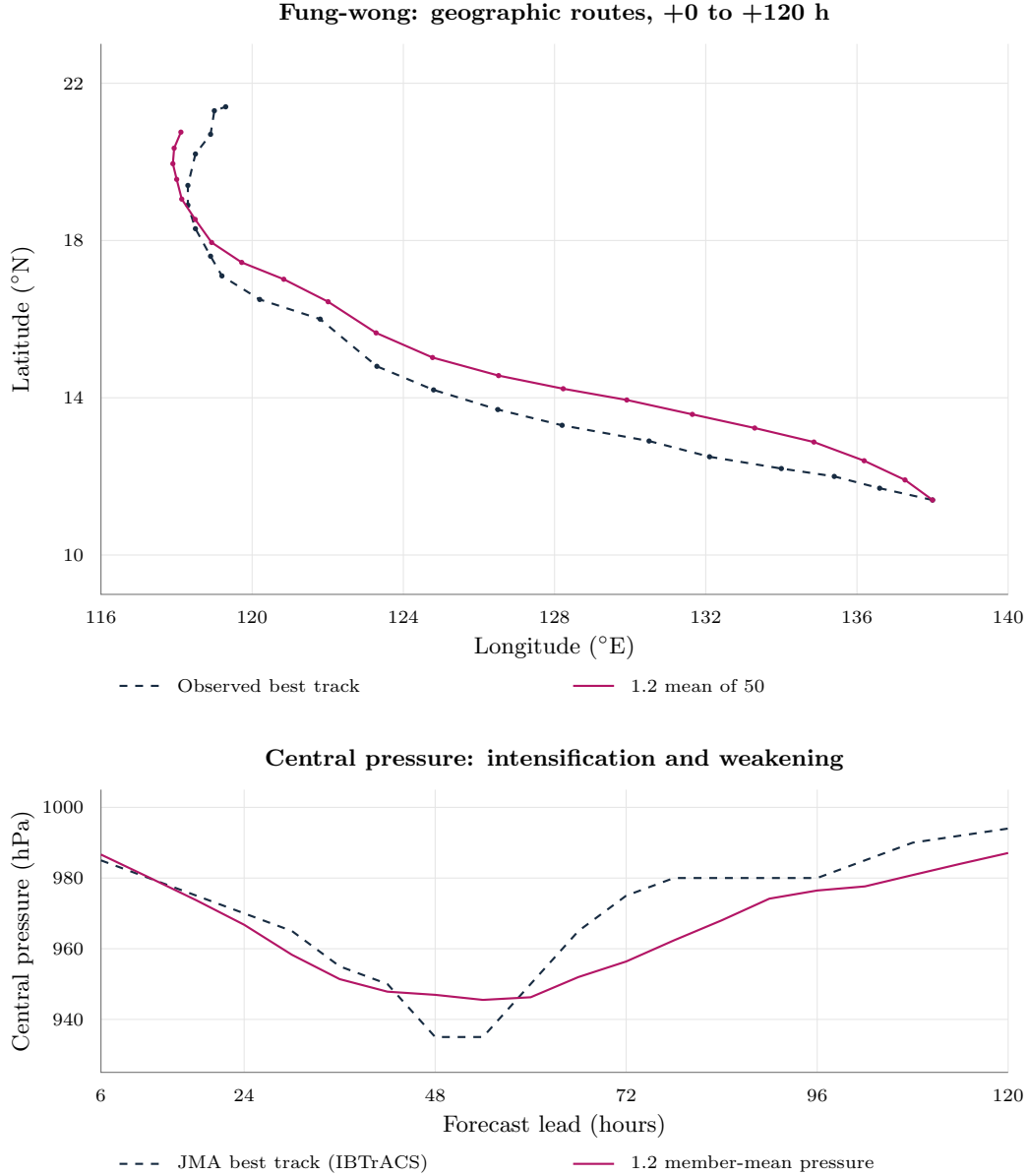


Figure 3: Selected development example, not representative performance. Markers on the route denote matching six-hour valid times, including the issue centre. Observations are verification overlays, never forecast inputs. Pressure MAE is 7.36 hPa over twenty valid labels. The broad track and pressure evolution are similar, but the curves do not perfectly coincide; peak intensity and weakening timing remain imperfect. The companion MP4 shows the actual common-grid model pressure fields and 4-hPa isobars; none of the pressure fields is moved to match observations.

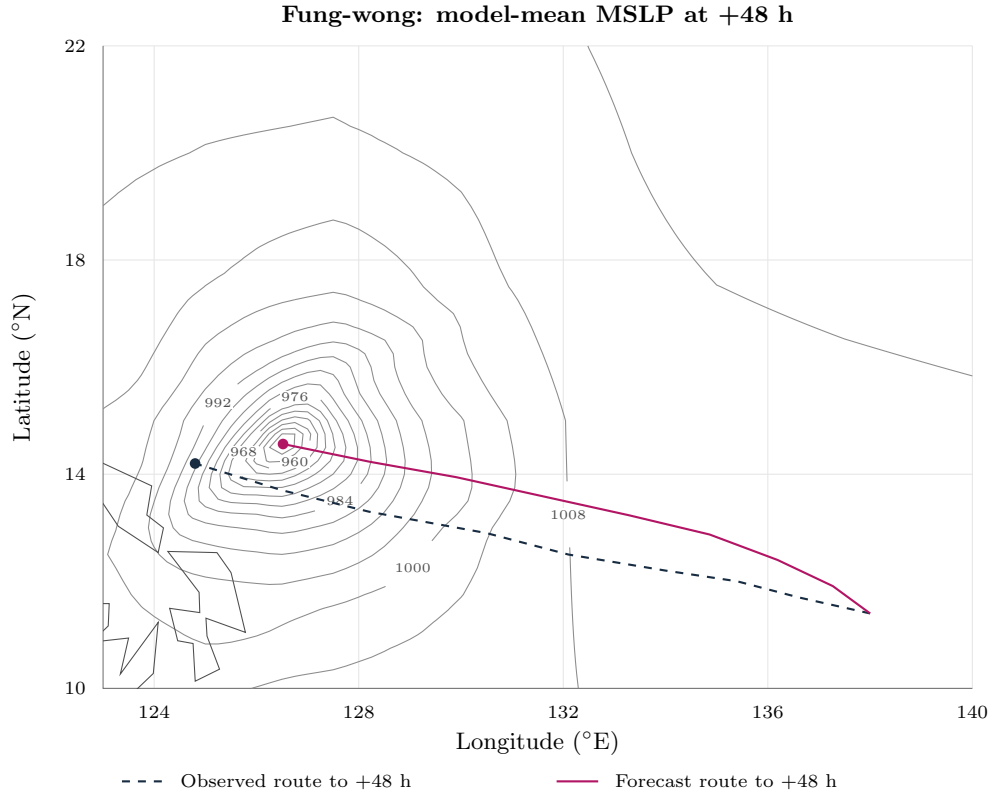


Figure 4: Unmodified saved physical ensemble-mean regional MSLP at +48 h (9 November 2025, 00 UTC), contoured every 4 hPa and labelled every 8 hPa. The source grid is sampled at 0.25°; this is not a claim of native resolution. Lines overlay the observed and forecast paths only through this valid time. The figure uses the portion inside the fixed regional patch; its western edge is 123°E. The map has not been shifted to match the observed centre, and its minimum need not equal member-mean central pressure.

10.1 Pressure and coverage limitations

Observed pressure is JMA best track as stored in IBTrACS TOKYO_PRES, not a JMA forecast. Native-detail input history was unavailable for this issue; the regional pressure field is a learned basin-plus-core reconstruction sampled every 0.25° , not a native-resolution forecast. Coverage masks and fixed regional boundaries must be retained. From +66 h the forecast centre leaves the fixed regional patch; the video then shows only the saved coarse basin field at that location. The moving-core readout still supplies central pressure; no detailed map core is invented outside coverage. The mean route and mean central pressure need not be the minimum of the common-grid mean field.

No verified same-270 1.1 pressure-prediction array is available in the route artifacts, so a pressure-comparison bar is not invented. On the separate, repeatedly inspected TIP ten-start diagnostic, central-pressure MAE is worse for 1.2 (30.7 versus 20.1 hPa). Overlapping issues of one storm are not ten independent tests. The route improvement cannot be read as an all-metric win.

11 Reproducibility and limitations

The figures above embed numeric values from the published saved artifacts, so this source compiles without external images. Model code, checkpoint provenance, plot data, selection rules and reproduction utilities are available in the release repository¹. The 270-case scores are in `evaluation/trackformer_1_2_vs_1_1_270_metrics.json`; Fung-wong coordinates, pressure, source hashes and member policy are in `evaluation/release_data/fung_wong_video.npz` and its JSON manifest. The [Fung-wong MP4](#) is an illustration of saved predictions, not a new inference experiment.

A separate expanded protocol freezes 1,473 daily issues from 270 distinct storms: average leads within each issue, issues within each storm, then storms equally. It is not the completed 270-case comparison reported here; its partial results are not used in these figures. Stronger generalization claims require a genuinely unused whole-storm holdout, fitting-only preprocessing, matched data availability and lead masks, and uncertainty estimates based on resampling whole storms rather than overlapping issues.

12 Conclusion

Trackformer 1.2 couples recurrent environmental evolution with a translating pressure anomaly and pressure-associated readout. Development results improve aggregate route measures relative to the archived 1.1 pipeline, while leaving large absolute errors, short-lead regressions and intensity failures. The architecture is a research model, not a substitute for official warnings; selected visually aligned routes do not establish operational reliability.

¹<https://github.com/yu314-coder/typhoon-predict>